Inverse Proportion

Exam code: 0607

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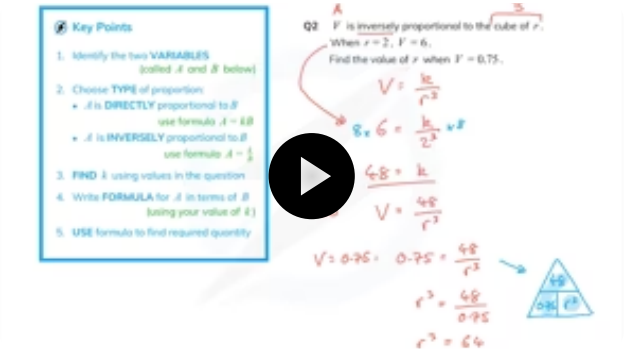

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Inverse proportion

What is inverse proportion?

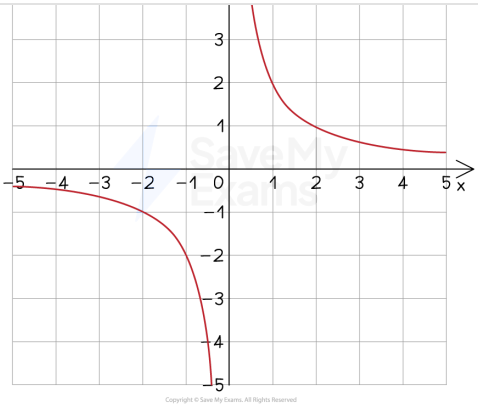
- **Inverse** proportion means as **one variable goes up** the **other goes down** by the same **factor**
  - If two quantities are **inversely proportional**, then we can say that one is **directly proportional** to the **reciprocal** of the other
- The symbol, $\propto$, is used to show that one quantity is "directly proportional to the reciprocal of" (inversely proportional to) another quantity
  - "y is inversely proportional to x" is written $y \propto \frac{1}{x}$
- If x and y are inversely proportional then
  - $\frac{1}{x} : y$ will always be the same
  - there will be some value of k such that $y = \frac{k}{x}$
- The **graph** of $y = \frac{k}{x}$ is shown below

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Example of a graph showing inverse proportion

How do I use inverse proportion with powers and roots?

- Problems may involve a variable being **inversely proportional** to a **power or root** of another variable
- For example
  - *y* is **inversely** proportional to the **square of *x***
    - $y \propto \frac{1}{x^2}$
    - means that $y = \frac{k}{x^2}$
  - *y* is **inversely** proportional to the **square root of *x***
    - $y \propto \frac{1}{\sqrt{x}}$
    - means that $y = \frac{k}{\sqrt{x}}$
  - *y* is **inversely** proportional to the **cube of *x***
    - $y \propto \frac{1}{x^3}$
    - means that $y = \frac{k}{x^3}$
  - *y* is **inversely** proportional to the **cube root of *x***
    - $y \propto \frac{1}{\sqrt[3]{x}}$
    - means that $y = \frac{k}{\sqrt[3]{x}}$
  - Each of these would have a **different type of graph**, depending on the power or root

How do I find the equation between two inversely proportional variables?

- Inverse proportion questions always have the same process:
  - **STEP 1**
    Identify the two variables and write down the **formula in terms of *k***
    - E.g. ***y*** is **inversely** proportional to ***x***

Find *k* by substituting any given values **from the question** into your formula, then solving to get *k*

- E.g. if you are told $y = 5$ when $x = 6$
- then $5 = \frac{k}{6}$ giving $k = 30$

- **STEP 3**
  Rewrite the formula with the **known value of *k*** from above (substitute it in)
  - $y = \frac{30}{x}$
  - This is the **equation** relating the two variables

- **STEP 4**
  Use the equation to answer other parts of the question
  - E.g. find y when $x = 2$
  - $y = \frac{30}{x}$ gives $y = \frac{30}{2} = 15$

Worked Example

The time, t hours, it takes to complete a project is inversely proportional to the cube root of the number, n, of people working on it.

If 27 people work on the project, it takes 50 hours to complete.

(a) Find an equation connecting t and n.

Answer:

Identify the two variables

$$t, n$$

We are told this is **inverse** proportion to the cube root of n
Write down the formula involving *k*

$$t = \frac{k}{\sqrt[3]{n}}$$

Find *k* by substituting in $n = 27$ and $t = 50$ (from the words in the question)

$$50 = \frac{k}{\sqrt[3]{27}}$$
$$50 = \frac{k}{3}$$
$$50 \times 3 = k$$
$$150 = k$$

Substitute this value of *k* back into the formula to get the full equation

$$t = \frac{150}{\sqrt[3]{n}}$$

Answer:

Use the formula to find n when $t = 60$

$$60 = \frac{150}{\sqrt[3]{n}}$$

$$\sqrt[3]{n} = \frac{150}{60} = 2.5$$

$$n = 2.5^3 = 15.625$$

A sensible answer here is a whole number (as n is the number of people)
15 people would not complete it in time, but 16 would

16 people

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